

When θ is 55° , $r_2 = 0$ (smallest), $r_1 = r$ (largest).

2 C $v_y^2 = u_y^2 - 2gy$

At the highest point, $v_y = 0$. Hence,

$$0 = u_y^2 - 2gy$$

$$u_y = \sqrt{2gy}$$

Since the maximum height reached is the same for all three paths, u_y is the same too.

Option A: $S_y = u_y t + at^2$, since all three paths end on ground level, time of flight is the same.

Path Z has the longest range. $S_x = u_x t$, path Z has the highest horizontal component and the largest initial speed ($u = \sqrt{u_x^2 + u_y^2}$).

Option B: $S_y = u_y t + at^2$, since all three paths end on ground level, time of flight is the same.

Option D: Path X has the shortest range. $S_x = u_x t$, path X has the lowest horizontal component.

3 A By Newton's second law, taking the direction of F_1 as positive